PORTFOLIO OPTIMISATION · DERIVATIVES & STRUCTURED PRODUCTS

Beyond Mean-Variance

Worked Examples

Three engines · one behavioural framework — real runs, with analysis

Sami Jeddou · behavioral-portfolio-optimizer.streamlit.app

HOW TO READ THIS

The framework, in one minute

Classical optimisers judge a portfolio by mean and variance alone. This framework adds a behavioural, mental-accounts objective — maximise expected return subject to a **downside constraint** — and lets you hold derivatives and structured products *inside* the portfolio. The three engines explore it from different angles, all on the same Das–Statman base case.

Grid — exact

Das–Statman's exact algorithm: a fine weight search under a VaR limit. Precise and definitive, for small problems.

Scalable — MC + CVaR

Monte-Carlo scenarios + a CVaR linear program. Scales to many assets and derivatives, with a coherent tail measure.

Back-test

Build on one window, hold blind on the next. Realised vs expected, plus alpha & beta against a benchmark.

Each example that follows shows the **setup**, the **result**, and a short analysis of *why* it comes out that way.

THE APPLICATION

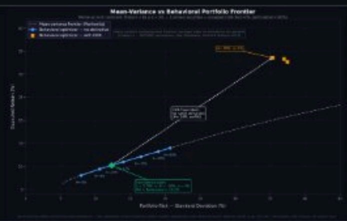
The app — at a glance

Portfolio Optimisation *with* Derivatives & Structured Products

Beyond Mean-Variance

Mental Accounting Framework

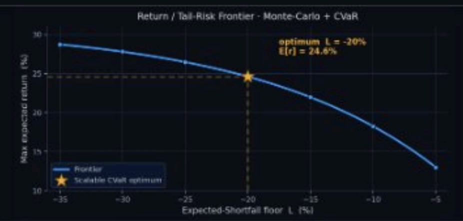
TOOLS



Grid Portfolio Optimiser

Exact grid engine on the Das–Statman states — VaR, thesis-faithful ES and rigorous-ES, with derivatives.

`grid · exact`



Scalable Portfolio Optimiser

Monte-Carlo scenarios + α-CVaR linear program — scales to large, multi-derivative portfolios.

`scenario · LP · beta`

Backtest

Out-of-sample walk-forward of the **Optimiser's** portfolios, derivative marked to market.

`↳ realised alpha & beta vs a benchmark`

`out-of-sample`

REFERENCE

About
Methods, framework and research.

Glossary AI-POWERED
VaR, ES, α-CVaR, copulas — plus natural-language Q&A.

User Guide
Step-by-step tour of the app (PDF download).

Built by Sami Jeddou · About & contact · Connect on LinkedIn

THE LANDING SCREEN

Three optimisers across the top — **Grid** (exact), **Scalable** (Monte-Carlo + CVaR) and **Back-test** — with a reference shelf below (About, AI Glossary, User Guide). Everything that follows is one of these tools in action, on the same Das–Statman base case.

REFERENCE

An AI-powered glossary

📚 AI Glossary & Reference

Click any term below for an AI-generated explanation, or type your own question. Answers are tailored to the context of behavioural portfolio optimisation.

👇 💡 After clicking a term or submitting a question, **scroll down** to see the answer at the bottom of this page.

Derivatives & structured products

Put option	Call option	Safety collar (long put + short call)
Aggressive collar (long call + short put)	Straddle (long call + long put)	Strangle (long call + long put, diff strikes)
Capital-guaranteed note (CGN)	Barrier-M note	Bull call spread (long call + short higher call)
Bear put spread (long put + short lower put)	Long butterfly (calls)	Call condor
Reverse convertible (bond − short put)	Discount certificate (capped underlying)	Outperformance certificate (geared upside)
Digital option	Zero-coupon bond	

Ask your own question

Type a term or question

e.g. What is the difference between VaR and ES?

🤖 Ask AI

EXPLAIN ANYTHING, IN CONTEXT

Every instrument and risk term — from a capital-guaranteed note to α-CVaR — has an **on-demand, context-aware explanation**, tailored to behavioural portfolio optimisation. A free-text box answers any question (e.g. *"What's the difference between VaR and ES?"*), so the framework is self-documenting for any reader.

REFERENCE

About & resources

Sami Jeddou

Senior Financial Services Executive — Transformation, Risk & Capital Markets

Risk · Capital Markets · Post-Trade & Clearing · High-Value Payments · Quantitative Finance · Front-to-Back Delivery · Regulatory Programs

📍 Paris, France

🔗 LinkedIn | GitHub | sami.jeddou@protonmail.com

About this app

📄 Download the User Guide (PDF)

A step-by-step guide to using the app.

📄 Download the technical paper (PDF)

A short companion note — the work, the approaches, the mathematical framework and the validation.

Beyond Mean-Variance Portfolio Optimiser is an interactive research tool that builds goal-based portfolios which can include **derivatives and structured products** — something the classical Markowitz mean-variance framework does not handle.

Rather than trading return against variance, it **maximises expected return subject to a downside rule you set** — either a maximum probability of loss (VaR) or a maximum expected loss in the tail (Expected Shortfall) — and finds the portfolio, including any hedges, that best meets that goal.

It provides **two optimisers**: a **grid optimiser** — the thesis method, which searches exhaustively and is exact for small portfolios — and a **scalable Monte-Carlo + CVaR optimiser** that simulates thousands of scenarios to handle large, institutional-size portfolios. An **out-of-sample back-test** then holds the chosen portfolio on later, unseen data and reports its realised performance (alpha, beta and R^2 versus a benchmark).

WHERE TO GO DEEPER

The About page carries the author profile and contact, a plain-language description of the framework and the two optimisers, and downloads for the **step-by-step user guide (PDF)** and the **technical paper (PDF)** — the fuller write-up of the methods, the mathematics and the validation behind these runs.

GET STARTED

Run it in **one click** — PC or phone

The live app is hosted on Streamlit — no install, nothing to set up. Open it straight from your browser, on a computer or a phone, just by clicking the link. For the full picture a computer is best: the charts and tables are information-dense.

sami-jeddou-behavioral-portfolio-optimizer.streamlit.app

Run it **locally** — full speed

WHAT	A step-by-step guide to run the optimiser on your own machine — at **full speed**, with no cloud sleep or timeouts.
WHY	The heaviest, high-resolution runs can be interrupted on the free hosted tier — it sleeps when idle and can time out or reboot. **The High-precision grid run in this document took ~92 minutes** — exactly the kind of job best done locally.
WHERE	Quick-start guide (PDF), direct download: raw.githubusercontent.com/SamiJeddou/behavioral-portfolio-optimizer/main/Run_Locally_Guide.pdf

WORKED EXAMPLE

01

Grid Optimiser — Exact

Grid Portfolio Optimiser

Exact grid engine on the Das–Statman states — VaR, thesis-faithful ES and rigorous-ES, with derivatives.

`grid · exact`

ENGINE 1 · GRID OPTIMISER — EXACT

A worked example — the setup

The grid optimiser reproduces Das & Statman's exact algorithm — a fine search over portfolio weights that evaluates the full return distribution at each point, maximising expected return subject to a downside (VaR) constraint. Below is one complete run, with and without a derivative.

1

📂 PORTFOLIO DATA

✨ AI-powered: How these inputs are built

- (•) Default (3-asset sample case)
- ( ) Live market data (Yahoo Finance)
- ( ) Enter manually
- ( ) Upload CSV

✓ Default base case loaded — Means: 5%, 10%, 25% | Std devs: 5%, 20%, 50%

2

📊 DERIVATIVE / STRUCTURED PRODUCT

Straddle (long call + long put)

✨ AI-powered: What is this instrument?

Underlying security

Sec 3 — High risk

ENGINE 1 · GRID OPTIMISER

Result · the efficient frontiers

✅ Computation complete (Total execution time: 92m 23s) ▼ click to expand

📌 **Optimisation Parameters** (summary)

DATA SOURCE
Default

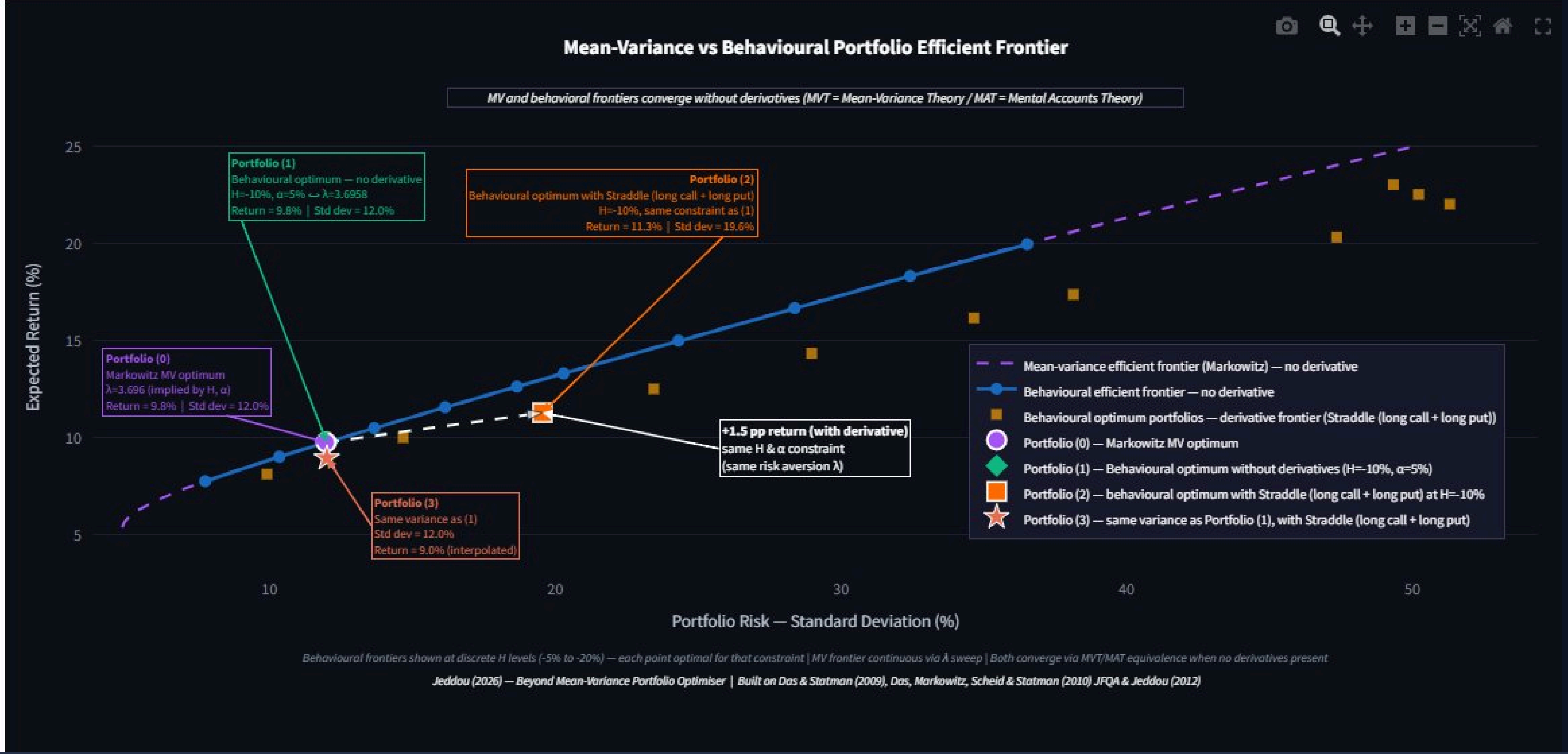
SECURITIES
Sec 1 — Low risk, Sec 2 — Mid risk, Sec 3 — High risk

DERIVATIVE
Straddle (long call + long put)
- Underlying: Sec 3 — High risk
- Maturity: 100.00%
- Strike: 100.00%

CONSTRAINT
VaR — H=-10%, α=5%

IMPLIED λ
3.6958

RESOLUTION
🎯 High precision



WHAT THE FRONTIER SHOWS

Without a derivative, the Markowitz and behavioural frontiers **coincide** — at this constraint the mental-accounts optimum is simply the Markowitz optimum (the MVT ≡ MAT equivalence).

Adding the straddle opens a **third frontier above both**: at the same -10% / 5% downside limit the optimal portfolio reaches a higher expected return (**+1.5pp**). It sits to the right — the gain comes with more variance.

Result · the four portfolios

Optimal portfolios with H=-10%, α=5% — implied λ = 3.6958

Summary — resulting portfolios

Portfolio	Expected return	Std deviation	Skewness	Δ vs (1)
• Portfolio (0) — Markowitz MV optimum (no derivative)	9.76%	11.99%	0.000	-0.00 pp
• Portfolio (1) — Behavioural optimum, no derivative	9.76%	12.01%	0.000	—
• Portfolio (2) — Behavioural optimum, with Straddle (long call + long put)	11.29%	19.56%	0.989	+1.53 pp
• Portfolio (3) — Same variance as Portfolio (1), with Straddle (long call + long put)	8.95%	12.01%	—	-0.81 pp

Δ vs (1) = expected-return gap relative to Portfolio (1). Portfolio (0) is a Gaussian mean-variance construct (skewness 0). Portfolio (3) is interpolated from the derivative frontier (indicative only).

Optimal portfolio (0) — Markowitz MV optimum (no derivative)

Minimum-variance portfolio at Portfolio (1)'s expected return — coincides with Portfolio (1) when it is mean-variance efficient (the MVT/MAT equivalence). Gaussian construct: skewness 0, tail probability (chance of finishing below H) from the normal model.

THE STORY IS IN THE SKEWNESS COLUMN

P0 (Markowitz) and P1 (behavioural, no derivative) are identical — 9.76% at ~12% vol, skewness 0. Without optionality the two frameworks agree.

P2 (with straddle) earns 11.29% — +1.53pp — but at 19.56% vol. It is mean-variance-*dominated*, yet meets the same downside limit: Markowitz rejects it, the mental-accounts objective accepts it.

Why? The straddle lifts **skewness from 0 to +0.99**. The uplift is paid in variance and collected in positive skewness — the dimension mean-variance can't see and the behavioural framework rewards.

P3 proves it: forced back to P1's variance, the straddle portfolio earns only 8.95% (**-0.81pp**). The edge needs the extra variance.

ENGINE 1 · GRID OPTIMISER

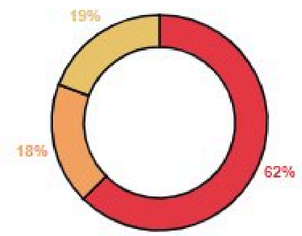
Inside · Portfolios 0 & 1 (no derivative)

Optimal portfolio (0) — Markowitz MV optimum (no derivative)

Minimum-variance portfolio at Portfolio (1)'s expected return — coincides with Portfolio (1) when it is mean-variance efficient (the MVT/MAT equivalence). Gaussian construct: skewness 0, tail probability (chance of finishing below H) from the normal model.

Expected return	Std deviation
9.76%	11.99%
Skewness	**Realised P(r<H)**
0.000	4.96%

Method: Markowitz mean-variance (SLSQP)



Portfolio weights

Sec 1 — Low risk — 62.5%

Sec 2 — Mid risk — 18.3%

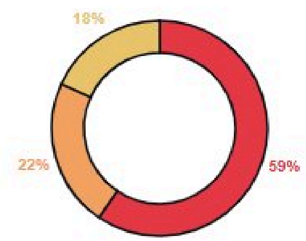
Sec 3 — High risk — 19.2%

Behavioural optimal portfolio (1) — no derivative

Maximises return subject to the downside constraint — reference portfolio (equivalent to Markowitz MV optimum)

Expected return	Std deviation
9.76%	12.01%
Skewness	**Realised P(r<H)**
0.000	5.00%
	✓ 5.00% ≤ α limit 5.00%

Method: Exhaustive grid search + COBYLA



Portfolio weights

Sec 1 — Low risk — 59.5%

Sec 2 — Mid risk — 22.3%

Sec 3 — High risk — 18.2%

NO-DERIVATIVE PORTFOLIOS

Both hold a similar mix (~60% low-risk, ~20% mid, ~19% high). P0 is the minimum-variance portfolio at P1's return; P1 maximises return under the -10% / 5% limit — and they **coincide**. The realised chance of breaching -10% is **4.96–5.00%**, right at the budget. Skewness is **0** — symmetric, Gaussian-like.

ENGINE 1 · GRID OPTIMISER

Inside · Portfolios 2 & 3 (with straddle)

Behavioural optimal portfolio (2) — with Straddle (long call + long put)

Same mental-accounting & risk-aversion constraint (H=-10%, α=5% ⟷ λ) — results may vary

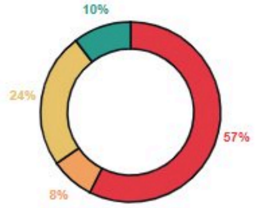
Expected return
11.29%
↑ +1.53pp vs portfolio (1)

Std deviation
19.56%

Skewness
0.989

Realised P(r<H)
4.99%
✓ 4.99% ≤ α limit 5.00%

Method: Exhaustive grid search + COBYLA



Portfolio weights
- Sec 1 — Low risk — 57.5%
- Sec 2 — Mid risk — 7.9%
- Sec 3 — High risk — 24.2%
- Derivative — 10.4%

At the same mental-accounting constraint (H=-10%, α=5% ⟷ λ), the optimum portfolio with Straddle (long call + long put) achieves 11.29% expected return vs **9.76%** for portfolio (1) without derivatives — a +1.53 pp gain (note: higher return may come with higher variance).

Optimal portfolio (3) — same variance as Portfolio (1), with Straddle (long call + long put) (interpolated)

Interpolated from the derivative frontier at the same std deviation as portfolio (1) — indicative only

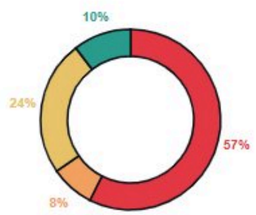
Expected return
8.95%
↓ -0.81pp vs portfolio (1)

Std deviation
12.01%

Skewness
0.989

Realised P(r<H)
4.99%

Method: Interpolated from derivative frontier — weights shown are from the closest optimised frontier point



Portfolio weights
- Sec 1 — Low risk — 57.5%
- Sec 2 — Mid risk — 7.9%
- Sec 3 — High risk — 24.2%
- Derivative — 10.4%

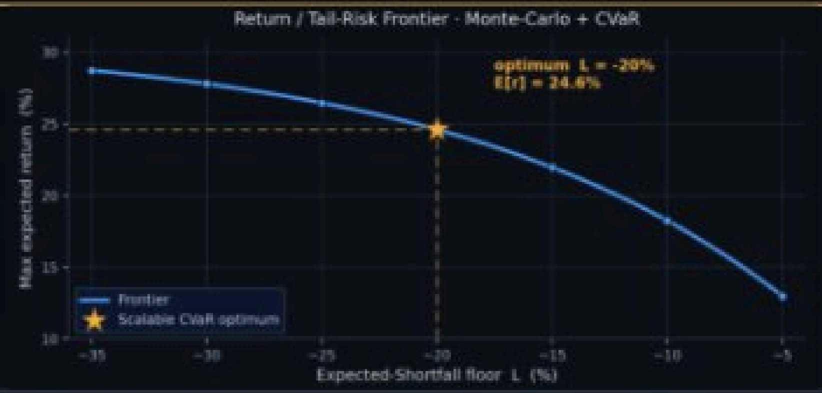
WITH THE STRADDLE

The optimal portfolio allocates **10.4% to the derivative** and trims the mid-risk sleeve, lifting expected return to **11.29% (+1.53pp)** while keeping the realised breach probability at **4.99%** — just inside the 5% limit (the constraint binds). Skewness jumps to **+0.99**. P3 holds the same weights but, read at P1's variance, sits lower (8.95%) — the same evidence that the straddle's edge is the skewness it buys, not free return.

WORKED EXAMPLE

02

Scalable Optimiser — Monte-Carlo + CVaR



[Scalable Portfolio Optimiser]

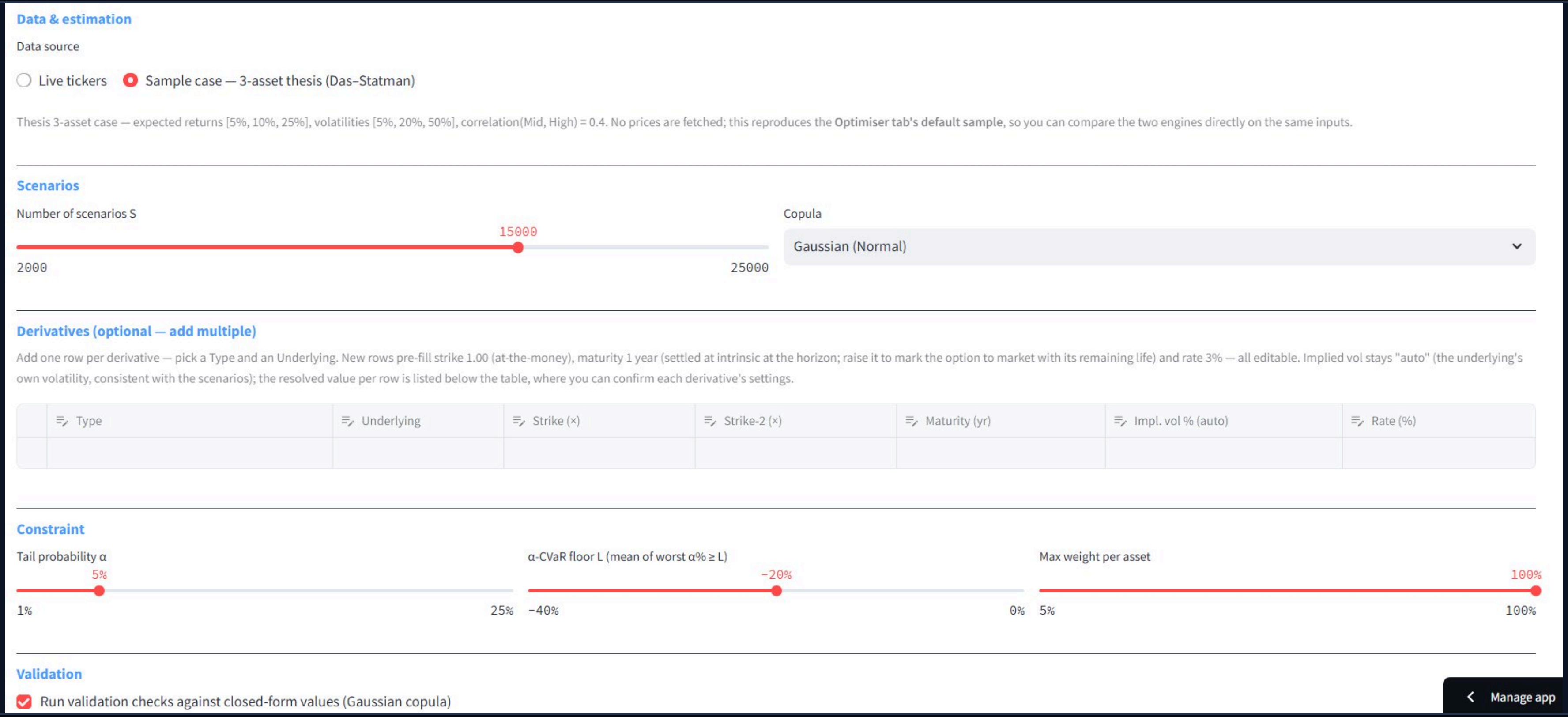
Monte-Carlo scenarios + α-CVaR linear program — scales to large, multi-derivative portfolios.

`scenario · LP · beta`

→

A worked example — the engine & setup

The scalable engine replaces the exact grid with a **Monte-Carlo + CVaR linear program** (Rockafellar–Uryasev). It draws thousands of joint scenarios from a copula on the same inputs, then maximises expected return subject to a *coherent* tail-risk floor — the average of the worst α% must stay above L. Cost is fixed in the number of scenarios, so it scales to many assets and derivatives where the grid's m^n state space cannot.



Same **3-asset thesis data** (returns 5/10/25%, vols 5/20/50%, ρ = 0.4) · **15,000 scenarios**, Gaussian copula · constraint: **α-CVaR floor L = -20%** at α = 5% · validation against closed-form Gaussian values on.

ENGINE 2 · SCALABLE OPTIMISER

Baseline · the CVaR optimum (no derivative)

Results

Expected return	Realised α-CVaR	Securities / derivatives
11.00%	-20.00%	3 / 0

★ **Scalable CVaR optimum — Monte-Carlo**

Expected return: 11.00% · **Volatility:** 15.19% · **Skewness:** 0.024

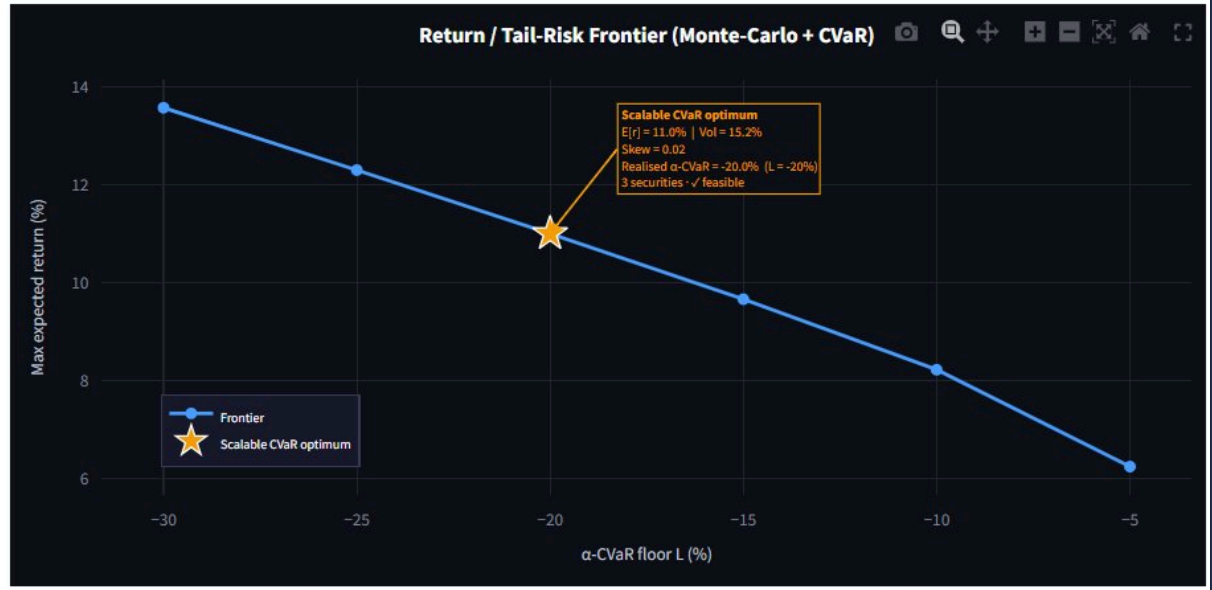
Realised α-CVaR (mean of worst α = 5%): -20.00% ✓ feasible — α-CVaR -20.00% ≥ L -20.00%

Universe: 3 securities

Scenarios: 15,000 (gaussian copula)

Objective: maximise E[r] subject to α-CVaR ≥ -20%

Approximate scenario-based optimum (CVaR linear program) — complements the exact grid engine.



⭐ marks the Scalable CVaR optimum (your resulting portfolio). Hover any point for its coordinates; drag to zoom.

BASELINE — NO DERIVATIVE

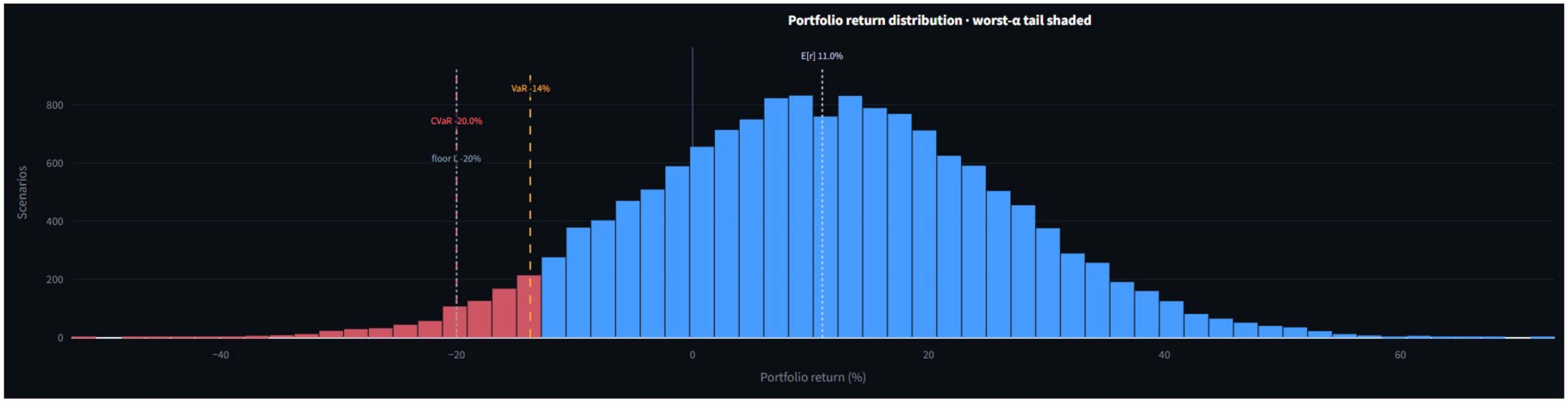
The optimum earns **11.00%** with the CVaR floor binding exactly (realised α-CVaR = -20.00%, feasible). Volatility 15.19%, skewness ≈ 0 — no optionality, so the distribution is symmetric.

Same data as the grid, different risk measure: the grid constrained VaR (≤5% chance of -10%) and chose 9.76%; the -20% CVaR floor is a looser leash, so here the optimiser reaches 11.00%. Different question, different answer — exactly why VaR and CVaR don't translate one-to-one. The frontier traces the trade-off: tighten the floor L (left) and achievable return falls.

What the CVaR floor controls

Portfolio return distribution

Simulated return of the chosen portfolio across every scenario. The shaded tail is the worst α; its average is the realised α-CVaR, held at or above the floor L — so the shaded region is exactly what the CVaR constraint controls.

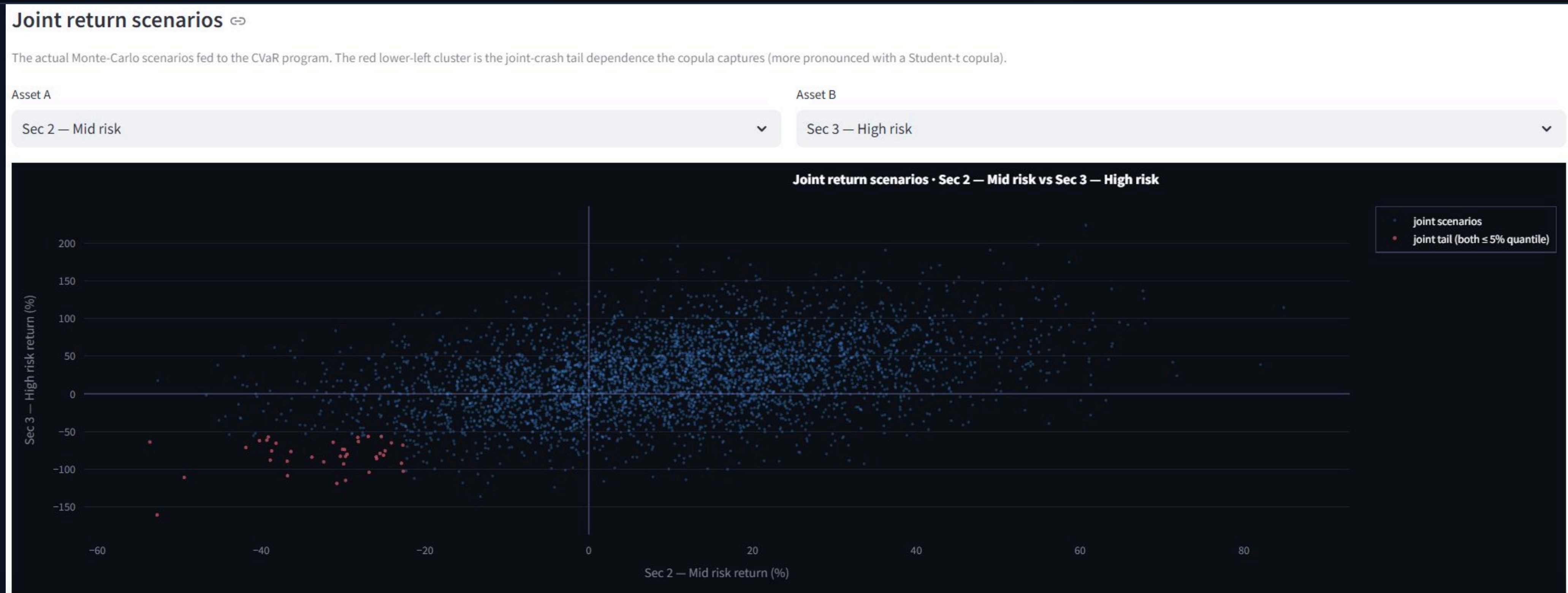


READING THE TAIL

The shaded worst-5% tail's *average* is the realised α-CVaR, held at the **-20% floor**. Note the two tail measures on the same portfolio: the 5% quantile (**VaR**) sits at **-14%**, while the **CVaR** (mean of the worst 5%) is **-20%**.

That -14% vs -20% gap is, in one picture, why the two measures aren't interchangeable: VaR marks where the tail begins, CVaR averages how bad it gets beyond. The program optimises the latter — the coherent one.

Under the hood · copula scenarios & weights



SCENARIOS AND THE BASELINE MIX

These are the actual scenarios fed to the program. The red lower-left cluster is the **joint-crash tail dependence** the copula captures (a Student-t copula would make it heavier) — dependence the closed-form grid can't model. The baseline optimum holds **50.9% low / 24.0% mid / 25.1% high**: a conventional spread, since without a hedge the CVaR floor caps how much of the volatile asset it can hold.

ENGINE 2 · SCALABLE OPTIMISER

Adding a protective put — +4.73pp

Results

Derivative pricing used (1)

- **Put·Sec 3 — High risk** — strike 0.80×, maturity 1.00 y, vol 50.0% (auto — Sec 3 — High risk σ), rate 3.0%

Expected return	Realised α-CVaR	Securities / derivatives
15.73%	-20.00%	3 / 1

★ Scalable CVaR optimum — Monte-Carlo

Expected return: 15.73% · **Volatility:** 29.44% · **Skewness:** 0.708

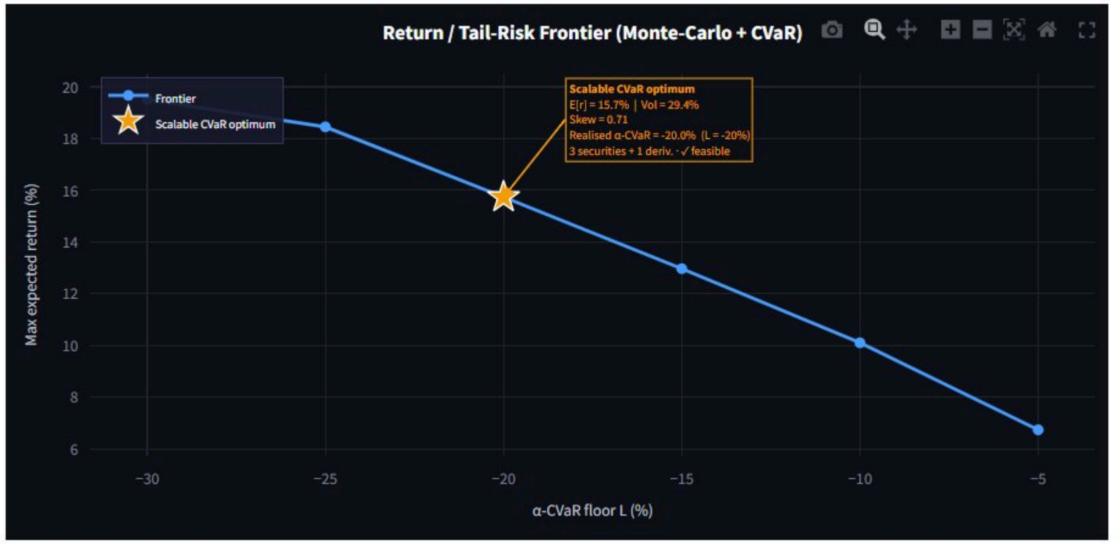
Realised α-CVaR (mean of worst α = 5%): -20.00% ✓ feasible — α-CVaR -20.00% ≥ L -20.00%

Universe: 3 securities + 1 derivative

Scenarios: 15,000 (gaussian copula)

Objective: maximise E[r] subject to α-CVaR ≥ -20%

Approximate scenario-based optimum (CVaR linear program) — complements the exact grid engine.



⭐ marks the Scalable CVaR optimum (your resulting portfolio). Hover any point for its coordinates; drag to zoom.

ONE DERIVATIVE CHANGES EVERYTHING

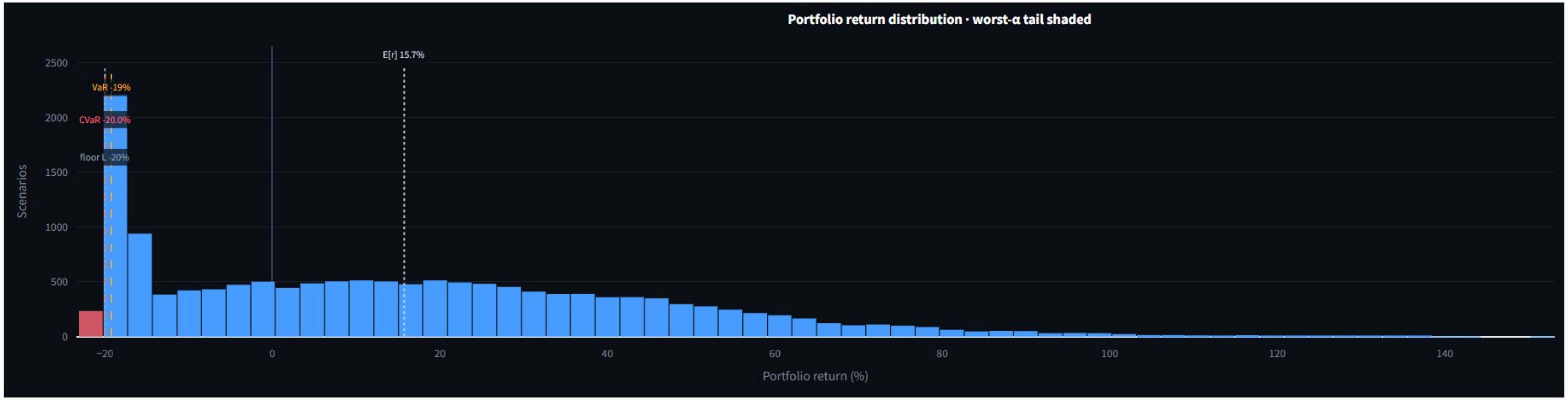
Add a put on the high-risk asset, struck **0.80×** (20% out-of-the-money), 1-year. At the **same -20% CVaR floor**, expected return jumps to **15.73% — a +4.73pp gain** over the no-derivative optimum. The put truncates the high-risk asset's left tail, so the CVaR constraint no longer caps the position — the optimiser chases the high-return asset and hedges the downside.

The signature: volatility 15.2% → **29.4%**, skewness 0.02 → **0.71**. Same floor, far higher return — paid for in (now mostly upside) variance and collected in positive skewness.

The put reshapes the distribution

Portfolio return distribution

Simulated return of the chosen portfolio across every scenario. The shaded tail is the worst α; its average is the realised α-CVaR, held at or above the floor L — so the shaded region is exactly what the CVaR constraint controls.



BOUNDED DOWNSIDE, OPEN UPSIDE

A wall of mass at the **−20% floor** (the put caps losses there) and a long right tail past +100% — the protective-put shape, and the source of the +0.71 skew. The weights show how: **69.9% high-risk** (up from 25%), 20.8% low, 3.1% mid, and a **6.2% put** doing the hedging. A 6% option allocation unlocks a near-tripling of the high-return position while holding the same tail-risk floor.

WORKED EXAMPLE

03

Out-of-Sample Back-test

Backtest

Out-of-sample walk-forward of the **Optimiser's** portfolios, derivative marked to market.

`↳ realised alpha & beta vs a benchmark`

`out-of-sample`

A worked example — build, then hold blind

The back-test is the out-of-sample check. It builds the optimal portfolio on one window, then **holds it fixed** — unchanged weights, buy-and-hold — through a later window it never saw, and measures what actually happened against what the model expected.

Inputs

Securities & settings

Tickers (comma-separated)

AAPL, MSFT, JPM

Return frequency

Daily

Construction period

the portfolio is built using data from this period

From

2012/01/01

To

2016/12/31

Evaluation period

the fixed portfolio is then held and measured over this period

From

2017/01/01

To

2017/12/31

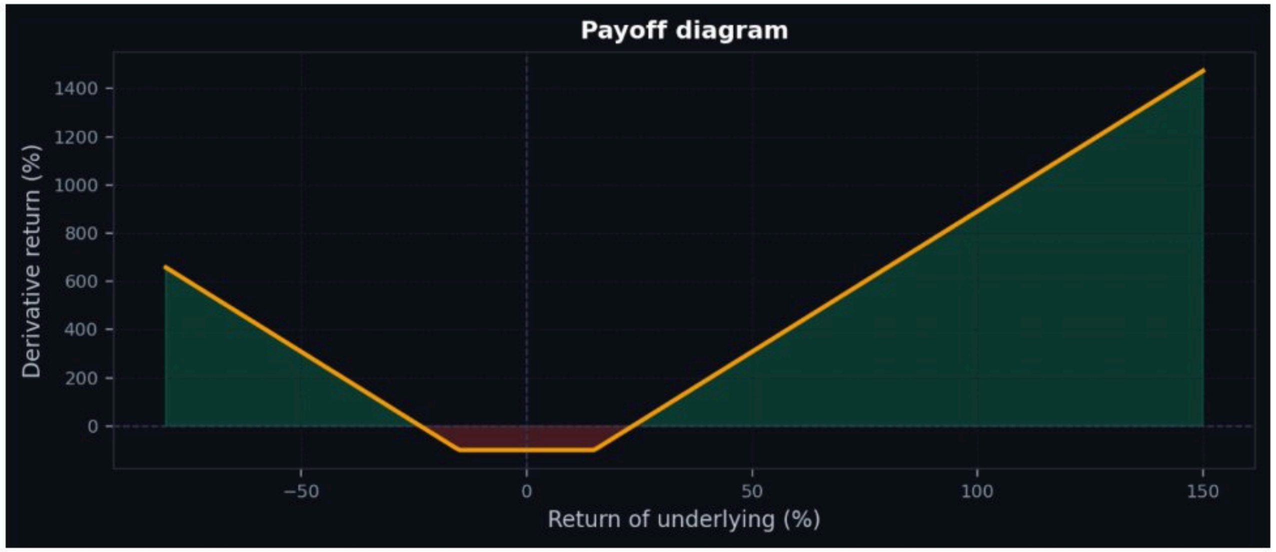
AAPL, MSFT, JPM (daily) · **built on 2012–2016**, then **held blind through 2017** · VaR constraint **H = −10%, α = 5%** · benchmark **S&P 500**, risk-free 3% · optional strangle overlay.

The overlay · a strangle

Derivative

Instrument

Strangle (long call + long put, diff strikes)



Underlying security (for the derivative)

Auto — highest volatility

Parameters

Put strike (×)

0.85

0.50 1.00

Call strike (×)

1.15

1.00 1.60

Illustrative payoff: derivative return vs underlying return (priced at ~25% vol). The backtest prices and marks the option with the construction-period volatility of the chosen underlying.

✨ AI-powered: What is this instrument?

THE DERIVATIVE

The optional overlay is a **strangle** — a long call (1.15×) and long put (0.85×) at different strikes, long volatility on both sides. The back-test prices and marks it with the *construction-period* volatility of the auto-selected underlying — **AAPL**, the highest-vol name (σ ≈ 25.7%). The optimiser's chosen weight in the derivative: **6.2%**.

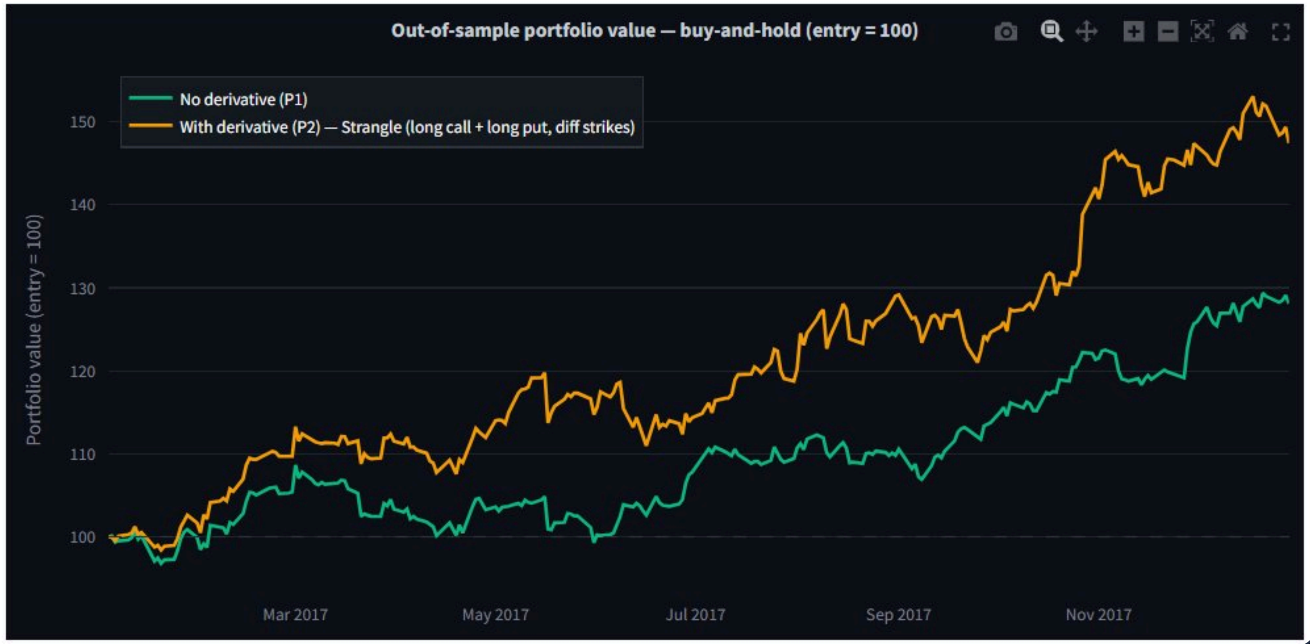
Result · held blind through 2017

Results

Underlying for the derivative: **AAPL** (highest volatility, σ = 25.7%). Optimal derivative weight in P2: **6.2%**. Option life T = 1.00y. Constraint: P(r < -10%) ≤ 5%.

Expected vs realised — out-of-sample

Metric	No derivative (P1)	With derivative (P2)
Expected annual return	23.31%	24.74%
Realised annual return	28.19%	47.59%
Expected annual volatility	20.66%	23.99%
Realised annual volatility	14.30%	18.61%
Model P(r	4.93%	5.00%
Window return (vs H = -10%)	28.08% ✓	47.39% ✓



WHAT ACTUALLY HAPPENED

Both portfolios did well, but the derivative portfolio (P2) far outpaced the plain one (P1): realised **47.59% vs 28.19%** — ~19pp more — with only a modest rise in realised volatility (14.3% → 18.6%). Neither breached -10% over the window.

Read it honestly: realised returns ran well above the model's expected ~23–25% because **2017 was a strong, trending year** for these names, and the strangle — long large moves — was rewarded. This is **one out-of-sample window**: a worked example of the workflow, not a claim of persistent edge.

ENGINE 3 · OUT-OF-SAMPLE BACK-TEST

Result · alpha, beta & verdict

Alpha / Beta — vs S&P 500 (^GSPC)

Realised over the evaluation window (250 daily periods), risk-free 3.00% p.a. β is the regression slope of excess returns, α the annualised Jensen intercept, R^2 its fit. Portfolio β is measured directly from the realised buy-and-hold path (so the derivative's non-linearity is captured).

Holding	β	Realised α (ann.)	R^2
AAPL	1.38	+18.42%	0.27
MSFT	1.31	+13.19%	0.34
JPM	1.49	-0.14%	0.37
Portfolio P1 (no derivative)	1.47	+1.99%	0.47
Portfolio P2 (with derivative)	2.10	+7.95%	0.56

Tip: enter an expected market return $E[R_m]$ in the inputs above to add a CAPM required-return and an ex-ante (expected) alpha column.

ALPHA, BETA & THE VERDICT

Against the S&P 500, the plain portfolio ran **β 1.47** with **+1.99%** realised alpha; the derivative portfolio ran **β 2.10** with **+7.95%** alpha. The strangle added both market exposure and genuine non-linear return — and β is measured directly from the realised buy-and-hold path, so the option's non-linearity is captured, not assumed.

Verdict: in this window the derivative-enhanced portfolio delivered more return and more alpha while keeping the -10% downside intact — consistent with the in-sample story. But it is a single favourable year and a high-β book; the honest takeaway is the **discipline the engine enforces** — build, freeze, and measure realised against expected — not the headline 47%.

IN SUMMARY

What these runs **show**

1. Derivatives earn their place by **reshaping the distribution** — adding positive skewness, truncating tails — which mean-variance can't see and the behavioural objective rewards. The grid's straddle lifted skewness 0 → 0.99 (+1.5pp); the scalable engine's protective put unlocked a far larger high-return position (+4.7pp).
2. **Same data, different risk measure.** VaR (grid, 9.76%) and CVaR (scalable, 11.00%) give different optima — neither wrong, just different questions about the tail.
3. **No free lunch.** The gains are paid for in variance; constrained to equal variance, the derivative edge disappears (grid P3, -0.81pp). The honesty is built in.
4. **Out-of-sample, discipline beats any single number.** Build, freeze, and measure realised against expected — that is the back-test's real contribution.

> **Try it —** live app · step-by-step user guide (PDF) · open-source code on GitHub. All links in the first comment.